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St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore – 607001
B.Sc Degree Examination – APRIL 2021

APH301 – ALLIED PHYSICS

Time : 3 hrs **Max Marks :75**

SECTION – A (10X2=20)

Answer **ALL** Questions

1. Write the Important characteristics of wave motion.
2. Sound travels faster on rainy days. Why?
3. Why automobile batteries have low internal resistance?
4. What is principle of a potentiometer?
5. Why the centre of the Newton's rings is dark?
6. Give the condition for sustained interference.
7. State the postulates of special theory of relativity.
8. What is the significance of the wave function?
9. Prove the Boolean identity (A+B) (A+C) = A+BC
10. Difference between analog signal and digital signal.

SECTION – B (5X5=25)

Answer **ALL** Questions

11. a) Drive Newton-Laplace formula for velocity of sound in gases.

(or)

b) Distinguish between intensity and loudness of sound.

12. a) Prove that the energy stored in a parallel plate capacitor is q2/2C.

(or)

b) Determine the internal resistance of a cell by using voltmeter.

13. a) Light of wavelength 600 nm falls normally on a thin wedge shaped film of refractive index 1.4, forming fringes that are 2mm apart. Find the angle of the wedges**.**

(or)

b) In Newton's ring experiment the diameter of certain order of dark rings measured to be double that of second ring. What is the order of the ring?

14. a) Derive Einstein's mass energy equivalence.

(or)

b) Derive an expression for de Broglie wavelength of matter waves.

15. a) State and prove De Morgan's theorems.

(or)

b) Describe the construction of Zener diode.

SECTION - C (3X10=30)

Answer any **THREE** Questions

16. Obtain an expression for the velocity of transverse wave in a stretched string. and When it is vibrating in fundamental mode.

17. Discuss the theory of interference in thin transparent film due to reflected light and obtain condition for the intensity to be maximum and minimum.

18. Expression for Schrodinger's equation. Indicate graphically wave function for such a particles.

19. Obtain an expression for the current flowing in a circuit containing a pure inductance.

20. Explain an experiment to determine the characteristics of a transistor in CE configuration. How the transistor parameters can be evaluated.

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